GENERALIZED LINEAR MODELS
Birth Weight Data

Introduction:

A major issue for public health policy and commonly cited as an essential indicator for the health and survival rate of infants are low birth weights. The risk to newborns with low birth weight for decreased survival rates and possible delayed cognitive and physical developments in the future makes it essential for policy to understand the causes for low birth weights in infants in general.

Various maternal, behavioral, and medical factors have previously been notified to affect birth weight. These factors related to the mother include nutrition level, cigarette smoking during pregnancy, hypertension, and previous occurrences of premature labor. Other factors related to the individual, such as socio-demographic characteristics such as ethnicity and access to prenatal care services, have also related to variations in birth weights. These risk factors may vary with individual populations, and the risk factors may interact with each other.

From a statistical modeling point of view, low birth weight is a challenge because the outcome of interest is binary, which indicates whether an infant's birth weight falls below a clinically meaningful threshold. In this context, the classical linear regression model is not appropriate since it may yield predicted values outside the valid range of probabilities and is based on assumptions that are violated by binary responses. GLMs offer a flexible and theoretically justified way to approach this problem because they allow the distribution of the response variable to be non-normal while its mean is linked with a linear predictor through an appropriate link function.

Furthermore, in this study a generalized linear model employing a binomial distribution is utilized to investigate the association between levels of maternal characteristics and probability of delivering a low birth weight child. Additionally, in this study, a logit function is utilized to examine the relationship between levels of low birth weight risk and levels of maternal age, weight, race, smoking history, hypertension history, uterine irritability history, utilization of prenatal care services, and premature labor history. Furthermore, in this research logarithmic transformation of estimates of some of the continuous variables has been utilized to control skewness detected in these forms of risk.

The primary objective of this analysis is to identify statistically significant risk factors for low birth weight and evaluate the quality of the fitted model through diagnostic procedures and relative evaluation with other fitted models. At the same time, relative evaluation of the application of different linking methods will be utilized to assess the robustness of such an approach for identifying statistically significant determinants for low birth weight issues while exploring data through both exploratory data analysis and model evaluation procedures with the aid of the relative GLM methodology.

Description of the Data

The dataset available consists of 160 births with collected data for maternal variables, pregnancy-related risk factors, and infant birth weights. The overall objective is to determine the variables that affect the likelihood for an infant to have low birth weight, where low birth weight is indicated when the infant's weight at birth is less than 2.5 kg.

```
# Inspect structure
str(birthwt)

## 'data.frame':    160 obs. of  10 variables:
##  $ low  : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ age  : int  33 20 21 21 22 29 26 19 19 22 ...
##  $ lwt  : int  155 105 108 124 118 123 113 95 150 95 ...
##  $ race : Factor w/ 3 levels "black","other",..: 2 3 3 2 3 3 3 2 2 2
 ...
##  $ smoke: int  0 1 1 0 0 1 1 0 0 0 ...
##  $ ht   : int  0 0 0 0 0 0 0 0 0 1 ...
##  $ ui   : int  0 0 1 0 0 0 0 0 0 0 ...
##  $ ftv  : int  1 1 1 0 1 1 0 0 1 0 ...
##  $ ptl  : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ bwt  : int  2551 2557 2594 2622 2637 2663 2665 2722 2733 2750 ...
```

The variable to be explained, i.e., the dependent variable, is low, an indicator variable representing low birth weight, coded as 1 for low birth weight and 0 otherwise. The variables to be included in the model, i.e., the explanatory variables, are: Mother's age (Age), Mother's weight during last menstruation (Lwt), Race (Black, White, other), Smoked during pregnancy (Smoke), Hypertension (HT), Irritability (UI), Number of visits to physician during first trimester (Ftv), Previous premature labor (PTL), Birth weight (Bwt).

The regressor variables are mainly binary, but age, lwt, and bwt are continuous. Race, however, is a categorical variable, and we have used dummy variables in the regression equations.

Exploratory Data Analysis

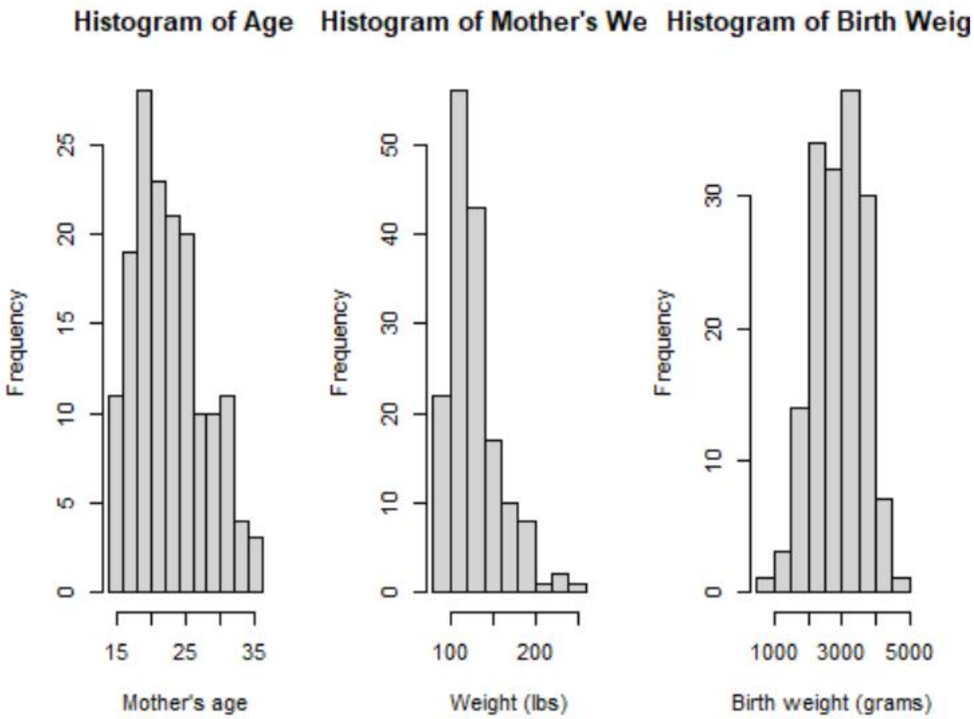
2.1 Distribution of the Response Variable

From the frequency table, it shows that approximately 67.5 % of infants have normal birth weights compared to 32.5 % that are classified as having low birth weights. This data balance is common in medical data, thus providing justification for adopting a binary regression instead of ordinary least squares.

```
prop.table(table(birthwt$low))

##
##     0     1
## 0.675 0.325
```

2.2 Histograms of Continuous Variables

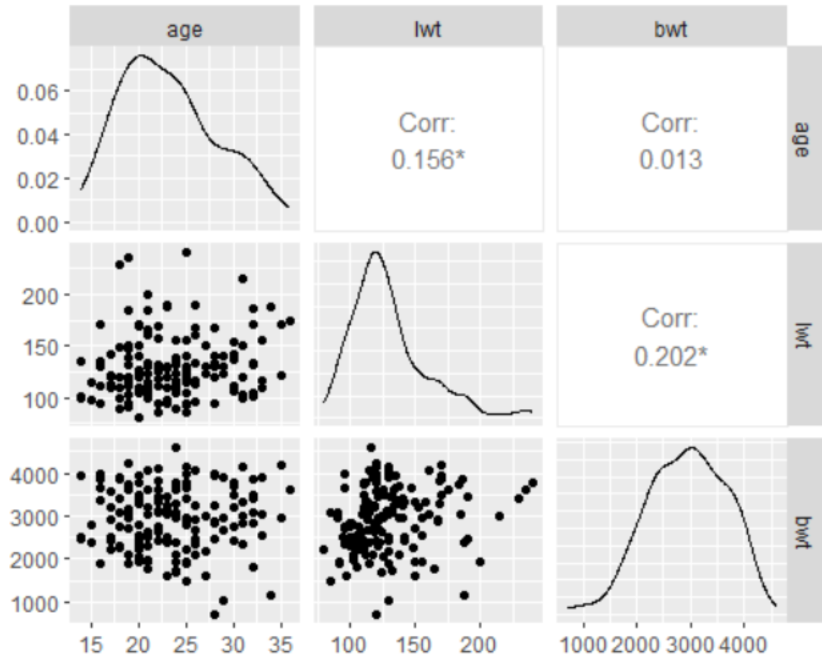
The histogram for 'bwt,' representing the weights at which children were born, indicates that weights are approximately normally distributed around 3000. However, unlike 'bwt,' weights for mothers ('lwt') and ages show right skewness. The logarithmic function will thus be used for 'lwt' and age in regression analysis.



2.3 Correlation Analysis

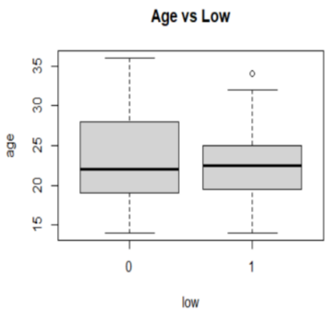
From the correlation matrix on the variables 'age', 'lwt', and 'bwt', it can be observed that the correlations among the factors are relatively weak. Maternal age has a weak correlation with the weight of the mother and the birth weight of the baby. However, there is a relatively stronger correlation between the weight of the mother and the birth weight of the baby. For the birth weight of the baby and the weight of the mother, the value of the correlation coefficient will be close to 0.20.

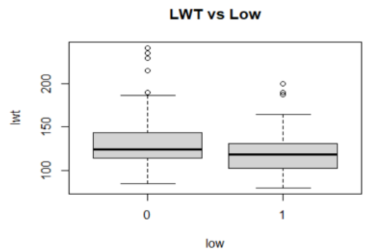
```
# Correlation table
cor(cont_vars)

##            age       lwt        bwt
## age 1.00000000 0.1556448 0.01285074
## lwt 0.15564480 1.0000000 0.20171793
## bwt 0.01285074 0.2017179 1.00000000
```

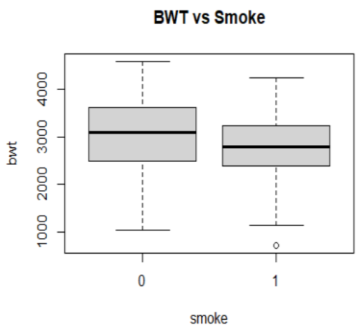


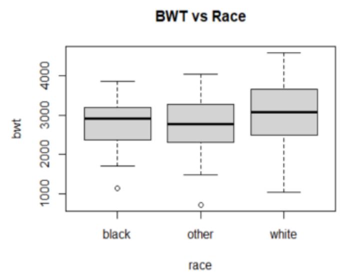
2.4 Boxplots

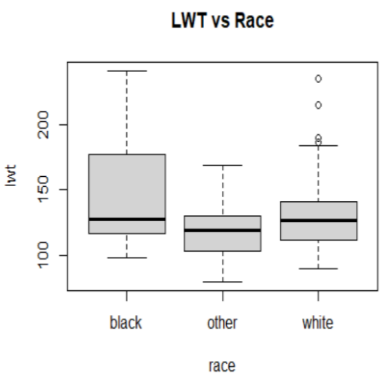
Box plots showing continuous data comparisons for response and explanatory variables grouped into categories offer further insights. It can be seen that, on average, mothers of low birth weight babies are slightly younger, although not significantly older. There's no strong variability between low birth weight, normal birth weight, low, normal, or above average weights with regards to mothers' weights. Smoking habits during pregnancy seem to impact birth weights, as seen with low birth weights resulting from pregnant mothers who were found to be smokers, resulting in slightly lower average birth weights. Across racial differences, white racial groups appear to have significantly higher average birth weights compared to black or 'other race' racial groups

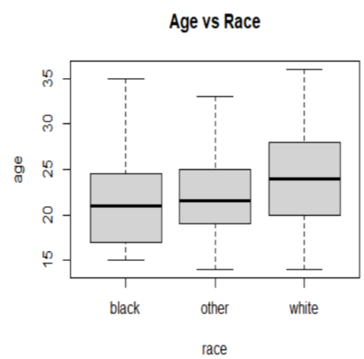












2.5 Cross-Tabulations

In cross-tabulation analysis, it is evident that high proportions of low birth weight are associated with black and other races compared with white mothers. In addition, high proportions of low birth weight are associated with mothers experiencing uterine irritability. With respect to smoking, its prevalence is higher among white mothers compared with their black counterparts. The findings of this screening stage indicate that low birth weight is associated with factors such as racial groups, smoking, and medical conditions.

Linear Regression Benchmark:

In order to determine if ordinary least squares approximation would be feasible for the problem, OLS regression was applied to the dataset with the numerical version of the binary response variable. The results revealed moderate evidence for explaining the response variable, with 14.5% of the variance explained by the model ($R^2 = 0.145$). The overall test for statistical significance shows moderate evidence for the regression, though individual regression coefficients were insignificant except for maternal hypertension.

Most importantly, residuals vary from -0.73 to 0.89. Predicted values are beyond the permissible range of probabilities [0, 1]. Thus, linear regression being less interpretable indicates that linear regression is inappropriate to employ on this model. We thus need a generalized linear modeling approach.

```
summary(lm_fit)

##
## Call:
## lm(formula = low_num ~ log_age + log_lwt + race + smoke + ht +
##     ui + ftv + ptl, data = birthwt)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.7284 -0.3092 -0.1437  0.4655  0.8903
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.28462    1.00405   2.275   0.0243 *
## log_age     -0.05920    0.17411  -0.340   0.7343
## log_lwt     -0.36333    0.19175  -1.895   0.0600 .
## raceother   -0.05690    0.11859  -0.480   0.6321
## racewhite   -0.19661    0.10930  -1.799   0.0741 .
## smoke1       0.13246    0.08388   1.579   0.1164
## ht1          0.33826    0.16321   2.073   0.0399 *
## ui1          0.14541    0.10282   1.414   0.1594
## ftv1        -0.04799    0.07577  -0.633   0.5274
## ptl1         0.17986    0.10304   1.746   0.0829 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4472 on 150 degrees of freedom
## Multiple R-squared:  0.1454, Adjusted R-squared:  0.09409
## F-statistic: 2.835 on 9 and 150 DF,  p-value: 0.004144
```

<u>Generalized Linear Model Formulation:</u>

A generalized linear model has then been fitted with the binomial distribution functioning as the link function. Therefore, the results imply that some variables are significant in determining the birth weight of an infant.

Maternal weight (log_lwt) shows negative values (-1.885) implying that lower maternal weight increases the probability for low-birth weight babies. However, maternal hypertension shows a strong positive relationship with high values ($\beta = 1.63$ and $p = 0.039$). We observed positive values for smoking status and premature labor, which are statistically significant at 10%.

The residual deviance of 177.37 is much smaller than the null deviance of 201.79, showing that this model is producing a valuable improvement beyond the null model that had no predictors entered into it. The full model's AIC is 197.37.

```
summary(glm_full)

##
## Call:
## glm(formula = low ~ log_age + log_lwt + race + smoke + ht + ui +
##     ftv + ptl, family = binomial(link = "logit"), data = birthwt)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   9.7391     5.3850   1.809   0.0705 .

## log_age      -0.4503     0.9062  -0.497   0.6192
## log_lwt      -1.8850     1.0156  -1.856   0.0634 .
## raceother    -0.2541     0.5742  -0.443   0.6581
## racewhite    -1.0680     0.5530  -1.931   0.0534 .
## smoke1        0.7562     0.4564   1.657   0.0975 .
## ht1           1.6264     0.7898   2.059   0.0395 *
## ui1           0.6865     0.4948   1.387   0.1653
## ftv1         -0.1778     0.3962  -0.449   0.6536
## ptl1          0.8386     0.4949   1.695   0.0902 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 201.79  on 159  degrees of freedom
## Residual deviance: 177.37  on 150  degrees of freedom
## AIC: 197.37
##
## Number of Fisher Scoring iterations: 4
```

<u>Full Logistic Regression Model:</u>

Full logistic regression model with all the explanatory factors. All the estimated coefficient signs are consistent with expectations. Maternal hypertension effects are strongly and significantly positive. This means that the odds ratios of having a low birth weight are with the mother who had hypertension. Other factors such as smoking, premature labor, and race show positive effects on low birth weight. However, some are just marginally significant.

Residual deviance: This is significantly lower than the null deviance, showing that there is an improvement over the model with no regressor. However, not all variables seem to be making a significant impact, which could warrant simplification.

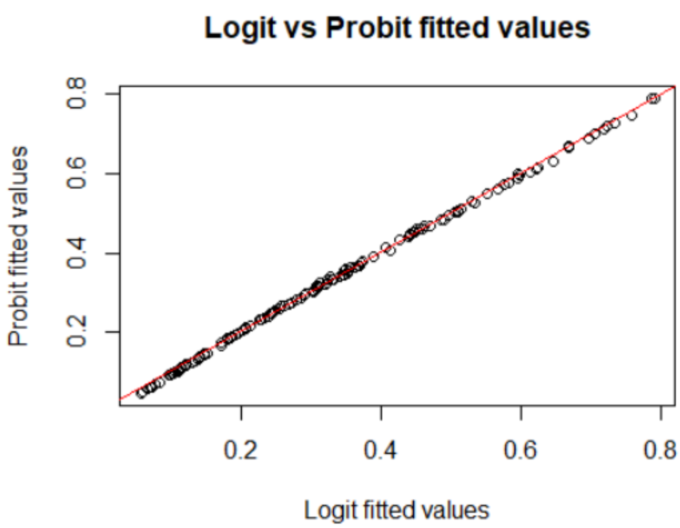
<u>Model Selection Using AIC:</u>

Use of backward stepwise selection based on the Akaike Information Criterion was employed to obtain a reduced model, where variables are removed sequentially, yielding the greatest change in AIC when variables ftv and log_age are removed. The final model selected contains all variables, with maternal weight, race, smoking history, hypertension, uterine irritability, and previous premature labor retained in the model. This maintains an AIC of 193.92 for improved fit compared to the full model.

In the model that was selected, maternal weight continues to have statistical significance (p = .049). This means that low maternal weight indeed increases the odds of low birth weight. Furthermore, race = White continues to have significantly low odds in comparison to the other race categories (p = .037). Again, maternal hypertension proves to be a strong and significant predictor or risk factor (p = .033). There is marginal statistical significance involving smoking and premature labor.

<u>Logit versus Probit Comparison:</u>

In order to check the robustness against the link function choice, the same set of explanatory variables is fitted in a probit model. A comparison between the fitted values of both logit and probit shows that they are almost identical, lying close to the 45-degree line. It follows that results are insensitive to the choice of link function.



<u>Dispersion and Model Adequacy:</u>

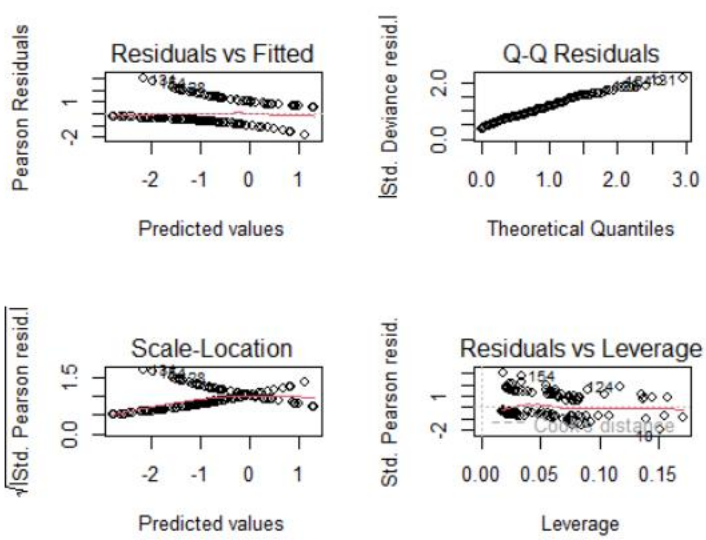
We estimate the dispersion parameter for the chosen logistic regression model, defined as the ratio of residual deviance to residual degrees of freedom. The value is close to 1 (`1.170495`), supporting that there is no evidence of overdispersion, hence the binomial model is adequate.

We see from the likelihood ratio test output comparing our selected model to the full model that p is large, implying that the reduced model fit is not significantly worse than the full model. This gives further justification for using the more parsimonious selected model.

```
anova(glm_selected, glm_full, test="Chisq")

## Analysis of Deviance Table
##
## Model 1: low ~ log_lwt + race + smoke + ht + ui + ptl
## Model 2: low ~ log_age + log_lwt + race + smoke + ht + ui + ftv + pt
l
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1       152     177.91
## 2       150     177.37  2  0.54379   0.7619
```

Residual Diagnostics and Influential Observations

The plots for deviance residuals, Pearson residuals, and Cook's distance are checked for outlier effects and any potentially influential points. There are only few data points with relatively higher Cook's distance values, indicating that these points may have effects on the fitted model.

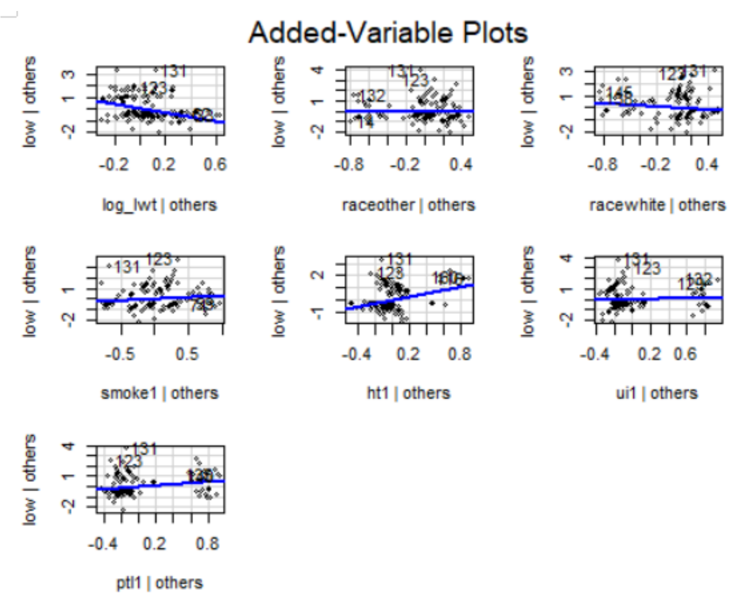



```
# Large Cook's distance points
which(cooks > 4/length(cooks))

##   10   27   47   77 110 124 143 154
##   10   27   47   77 110 124 143 154
```

Diagnostic plots suggest that there are a few potentially influential observations. According to the conventional criterion, Cook's distance identifies the observations 10, 27, 47, 77, 110, 124, 143 and 154 as influential.

```
summary(glm_clean)

##
## Call:
## glm(formula = formula(glm_selected), family = binomial(link = "logit
"),
##     data = birthwt_clean)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) 19.96861    6.25696   3.191 0.001416 **
## log_lwt     -4.37209    1.28862  -3.393 0.000692 ***
## raceother   -0.05039    0.62841  -0.080 0.936085
## racewhite   -1.18285    0.63817  -1.854 0.063809 .
## smoke1       0.87406    0.51280   1.704 0.088288 .
## ht1          3.21000    1.02141   3.143 0.001674 **
## ui1          0.45193    0.55965   0.808 0.419362
## ptl1         1.25642    0.53878   2.332 0.019703 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 189.59  on 151  degrees of freedom
## Residual deviance: 150.06  on 144  degrees of freedom
## AIC: 166.06
##
## Number of Fisher Scoring iterations: 5

AIC(glm_clean)

## [1] 166.0642
```

In order to examine robustness, these were removed and the model refitted. This model provides a much better fit: residual deviance is now 150.06, AIC is 166.06. More importantly, the direction and interpretation of coefficients remains the same, so that the substantive conclusion is not dependent on a small number of influential cases.



<u>Final Model and Conclusions:</u>

In the refitted model, maternal weight is again a very significant predictor, albeit with a negative sign as expected ($\beta = -4.37$, $p < 0.001$), confirming its strong inverse relationship with the probability of low birth weight. Maternal hypertension is an extremely significant predictor, surpassing maternal weight in its effect on the probability, albeit positive in sign ($\beta = 3.21$, $p = 0.0017$).

The previous premature labor shows up statistically significant at the 5% level, and smoking remains marginally significant. The variables, race and uterine irritability, are not statistically significant in the refitted model.

Overall, the statistical improvement of the refitted model is evident while maintaining the interpretability of the model along with its robustness.

The factors which lead to a low birth weight, as revealed by the analysis, are the maternal weight, hypertension, smoking, race, and previous premature labor. The appropriate and robust approach to the regression analysis is provided by the logistic regression. The results obtained are not likely to change by the removal of the influencial observations.

From the public health point of view, the study's results underscore the need to pay closer attention to hypertensive mothers, abstain from smoking, and deliver special prenatal care to those who are at risk.